

Fig. 2: Block diagram/Flowchart

BILL OF MATERIALS

Component	Value/Type	Quantity	Price (₹)
ESP32 board	IndusBoard (ESP32-S2) or any ESP32 board	1	1800
Temperature sensor	DS18B20, waterproof probe	3	180 (60 each)
Resistor	4.7kΩ, ¼ W, 5% carbon film	3	Miscellaneous 520
Ceramic disc capacitor	0.1μF	1	
USB cable	For programming and power	1	
Protective enclosure	For field use (optional)	1	
Total			**2500**

Note: *The temperature sensors remain embedded in the concrete and cannot be reused.*

and a plain-language interpretation.

Why does this matter? A simple thermometer shows only the present temperature. This device keeps a verifiable record of the whole curing period, helping check casting and curing practices on poorly supervised sites, where inadequate placement and curing have contributed to beam and bridge failures. Fig. 1 shows the author's prototype, while Fig. 2 presents the block diagram of the project.

What will you learn?

- How an ESP32 reads multiple DS18B20 sensors over a OneWire bus.
- How to compute temperature statistics and integrate temperature over time.
- Why concrete age and concrete maturity are not the same thing.
- How to serve a live dashboard from the ESP32 in access-point mode.
- How to export data as CSV, a graph, and an automatically generated report.

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READY TO BUILD IT YOURSELF?

Head to **https://efy.co.in/gigh53** for the complete circuit diagram, component list, PCB layout, step-by-step build instructions, troubleshooting tips and demo videos. Everything you need to take this project from idea to working prototype is waiting for you there!

